

UNIVERSITY NAME

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N-Puzzle Proposal

AI Project Proposal

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Project Overview

- **Project Name:** N-Puzzle Solver
- **Project Description:** This project is a desktop game that simulates the classic N-Puzzle (sliding puzzle) game, where the player must arrange the tiles into the correct numerical order. The game supports both manual play and automatic solving using the A* search algorithm with the Manhattan distance heuristic.
- **AI Algorithm:** A* Search Algorithm (with Manhattan Distance as a heuristic).
- **Project Type:** Desktop Application.
- **Used Technologies:**
 - Python.
 - Pygame.
 - A* Search Algorithm.
- **GitHub Link:**
<https://github.com/MohamedNabil114/N-puzzle-game>
- **YouTube Link:**
<https://youtu.be/HEoLoa1pUZU>
- **Image team Link:**
<https://postimg.cc/vxF3k66x>



Idea Explanation

Our project focuses on developing the classic "N-Puzzle" game, specifically the 8-puzzle version (3x3 grid).

The objective of the game is to move the tiles within the grid until they are ordered correctly, with one empty space.

The game will offer two modes:

- Manual Mode: The user manually solves the puzzle by clicking the tiles.
- Automatic Mode: The system solves the puzzle automatically using the A* (A-star) search algorithm.





Objectives

- Build an interactive intelligence-based game using Artificial Intelligence (AI) techniques.
- Apply search algorithms, particularly A^* , with the Manhattan distance heuristic.
- Practice problem formulation in AI.
- Develop an interactive graphical user interface (GUI) using the Pygame library.





PEAS

01 Performance

Minimize the number of moves needed to solve the puzzle, or minimize the time taken to solve it.

02 Environment

A 3x3 tile grid with numbered tiles and one empty space

03 Actuators

Movements of tiles (either by user input or automated solving).

04 Sensors

Current state of the grid (positions of tiles)



ODESDA

01 Fully Observable

The agent can fully observe the complete state of the puzzle (position of each tile).

02 Deterministic

Given a current state and a move (up, down, left, right), the next state is fully predictable.

03 Sequential

The current move affects future moves; it's a sequence of dependent steps to reach the goal.

04 Static

The environment (puzzle) does not change while the agent is deciding the next move.

05 Discrete

There are a finite, clearly defined number of moves (up, down, left, right).

06 Single Agent

Only one agent is solving the puzzle; there are no other competing agents.



Agent Type

Goal-based Agent

The agent acts to achieve a specific goal (correctly ordering the tiles) and makes decisions based on the current state and the desired goal. (In automatic solving mode, it behaves like a search-based agent.)

Problem Formulation

- **States:** Each state represents the arrangement of the tiles on the board.
- **Initial State:** A randomly shuffled, solvable configuration of tiles.
- **Actions:** Moving a tile adjacent to the empty space (up, down, left, or right)
- **Transition Model:** Defines how a tile move results in a new board state.
- **Goal Test:** Check if the board matches the goal configuration.
- **Path Cost:** Each move has a cost of 1; the objective is to minimize the number of moves.



N-Puzzle

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Thanks

